Bohao Peng

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EDUCATION

Harbin Institute of Technology, Shenzhen 2018 – Present
Bachelor of Engineering - Computer Science and Technology
GPA: 93.11/100, 3.97/4; Bonus Points for Contests: 5/5
Rank: 1/201

HONORS & AWARDS

- **National Scholarship** 2019
  - ✧ Top **1%**.
- **ACM – ICPC, International Collegiate Programming Contest, reginal: Gold Medal** 2020
  ACM – ICPC, International Collegiate Programming Contest, reginal: Gold Medal 2019
  ACM – ICPC, International Collegiate Programming Contest, reginal: Silver Medal 2018
  - ✧ Team leader.
  - ✧ Contribute first in the competition.
- **CCPC, China Collegiate Programming Contest (Guangdong): Gold Medal** 2021
  CCPC, China Collegiate Programming Contest: Silver Medal 2019
- **The Title of First Ten Top College Students in Harbin Institute of Technology** 2020
  - ✧ The highest honor for students.
  - ✧ The only student from the department of Computer Science and Technology.
- **Chunhui Innovation Achievement Scholarship** 2021
  - ✧ Top **0.1%**.
  - ✧ Reward for innovative invention in medical assistance systems.
- **The First-class Academic Scholarship** 2019&2020
- **Li Guang Scholarship** 2020
- **Outstanding Student in Harbin Institute of Technology** 2019&2020

RESEARCH EXPERIENCE

Sperm Morphology Analysis Based on Deep Learning and Shape Analysis 2019 – Present
Team leader, 15k+ lines of code

- National Invention Patent: *Jingyong Su, Bohao Peng, Kezhou li. A Method and Device for Sperm Morphology Analysis. Patent Number: 202110083620.6.*
  - ✧ Develop a system to help doctors diagnose, which can automatically locate the sperms and analyze their morphology.
  - ✧ Complete the system architecture and hardware driver design, algorithm development with Pytorch and interactive interface development with PyQt5.
- Paper: *Bohao Peng, Kezhou Li, Dongzhi Jiang, An Automatic Sperm Analysis Method based on Semantic Segmentation and Shape Analysis.*
  - ✧ Design a novel two-stage framework with res-net and unet++. After object detection and

sperm quality classification, semantic segmentation for contour detection and shape analysis are used to deal with the problems due to imbalance of positive and negative samples in the dataset.

Online Judge System for the Guangdong-Hong Kong-Macao Greater Bay Area Informatics Competition 2020 – 2021
Leader of live version, 3k+ lines of code

- The online judge system is used to test programs in programming contests. The system can compile and execute the competitor's code, then compare the output with standard output and return the result.
- Develop functions iteratively for live competitions, such as communication between competitors and competition judges, message notification, etc.

3D Subtomogram Alignment and Averaging 2021 – Present

- Cryo-electron tomography (cryo-ET) allows the 3D visualization of a small volume of vitreous biological sample with nanometer resolution.
- Geometric unsupervised matching model finds the geometric correspondence between two images, and spatial transform network is used for 3D subtomogram alignment and averaging.

TEACHING EXPERIENCE

Instructor of ACM – ICPC Training Team 2018 – Present

- Hold *China Collegiate Programming Contest (Final), 2018* as a student representative.
- Initiate and hold the first *Programming Contest of Harbin Institute of Technology (Shenzhen)* as the main person in charge.
- Teach courses includes:
  - ✧ Basic algorithm.
  - ✧ Advanced data structure.
  - ✧ Graph theory.
  - ✧ Discrete mathematics.

SKILLS & COURSES

Computer Skills:

- ✧ C, C++, Python, Kotlin, Javascript, Verilog, etc.
- ✧ Linux, Git, Qt, Spider, Nodejs, MySQL, Pytorch, etc.

Curriculum Highlights:

- ✧ Mathematics: Set Theory and Graph Theory, Probability Theory, Mathematical Logic, Computational Methods
- ✧ Computer Theories: Data Structures, Digital Logic Design, Design and Analysis of Algorithms, Principles of Computer Organization, Computer Networks, Operating System